

DSA0603 – Data Handling and Visualization

Slot B | CO2 – Assessment Test 4 (AT1)

Chart Construction Test

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Question 1 — Healthcare Analytics

Interdisciplinary Integration: Healthcare + Medical Statistics + Data Visualization

R Program (ggplot2)

```
library(ggplot2)

patients <- data.frame(
  Patient = c("P1", "P2", "P3", "P4", "P5", "P6", "P7", "P8"),
  Age = c(24, 32, 41, 48, 56, 65, 38, 29),
  BMI = c(19.8, 24.5, 28.1, 30.3, 33.2, 36.5, 26.2, 22.1),
  BloodPressure = c(110, 118, 132, 145, 152, 168, 124, 116),
  Severity = factor(c("Low", "Medium", "High", "High",
                     "Critical", "Critical", "Medium", "Low"),
                  levels = c("Low", "Medium", "High", "Critical"))
)

ggplot(patients, aes(x = Age, y = BloodPressure,
                    color = Severity, size = BMI)) +
  geom_point(alpha = 0.85, stroke = 0.6) +
  scale_color_manual(values = c("Low" = "#2ecc71",
                                "Medium" = "#f1c40f",
                                "High" = "#e67e22",
                                "Critical" = "#e74c3c")) +
  scale_size(range = c(3, 10)) +
  labs(title = "Age vs Blood Pressure by Disease Severity",
       x = "Age (years)", y = "Blood Pressure (mmHg)") +
  theme_minimal(base_size = 13)
```

Output

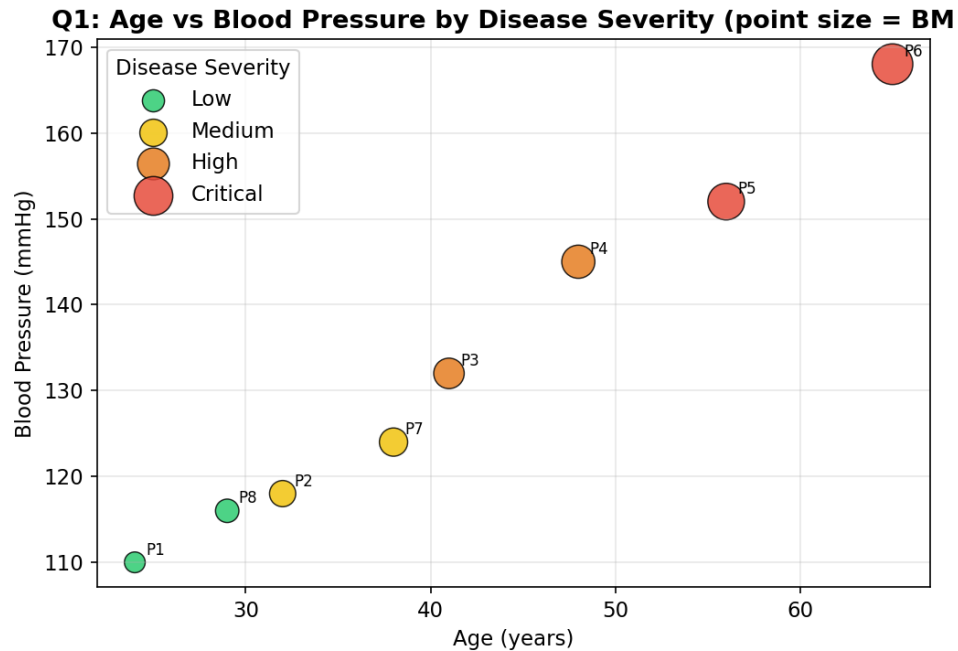


Figure 1: Age vs Blood Pressure, colored by Disease Severity, sized by BMI

Interpretation

- **Position mapping:** Age (x-axis) and Blood Pressure (y-axis) are continuous variables placed on Cartesian axes, which is the most natural mapping for showing a positive trend — as patients age, blood pressure tends to rise.
- **Color mapping:** Disease Severity is an ordinal categorical variable, so it is mapped to color using a traffic-light-style palette (green → yellow → orange → red). This exploits pre-attentive color perception so a clinician can instantly spot high-risk (red) patients.
- **Size mapping:** BMI is mapped to point size rather than color because it is a second continuous variable; using size avoids overloading the color channel and lets three variables (BP, Age, BMI) plus one categorical variable (Severity) be read from a single 2-D plot.
- **Why it communicates risk effectively:** Together, the largest, reddest points (e.g., P5, P6) instantly flag the oldest, heaviest, most critical patients — communicating compound health risk far faster than a table of numbers.

Question 2 — Climate Science

Interdisciplinary Integration: Environmental Science + Mathematics + Data Visualization

R Program (ggplot2) — Cartesian Line Chart (Temperature)

```
library(ggplot2)

climate <- data.frame(
  Month = factor(c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                  "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
                levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                          "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")),
  Solar = c(180, 200, 240, 280, 320, 300, 260, 250, 240, 220, 190, 170),
  Temperature = c(20, 23, 29, 35, 39, 36, 32, 31, 30, 28, 24, 21),
  Rainfall = c(25, 18, 12, 8, 5, 65, 150, 170, 120, 85, 45, 30)
)

ggplot(climate, aes(x = Month, y = Temperature, group = 1)) +
  geom_line(color = "#e74c3c", size = 1.1) +
  geom_point(color = "#e74c3c", size = 2.5) +
  labs(title = "Monthly Temperature Variation",
       x = "Month", y = "Temperature (°C)") +
  theme_minimal(base_size = 13)
```

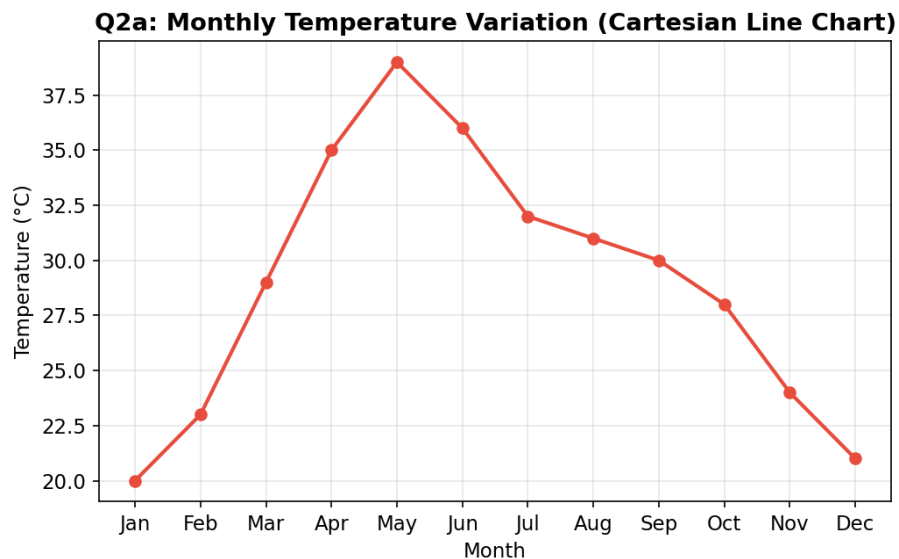


Figure 2a: Cartesian line chart of monthly temperature

R Program (ggplot2) — Polar Coordinate Chart (Rainfall)

```
ggplot(climate, aes(x = Month, y = Rainfall, fill = Rainfall)) +
  geom_col(width = 0.9, color = "black") +
  scale_fill_gradient(low = "#deebf7", high = "#08519c") +
  coord_polar() +
  labs(title = "Annual Rainfall (Polar Chart)", x = "", y = "Rainfall (mm)") +
  theme_minimal(base_size = 13)
```

Q2b: Annual Rainfall (Polar Coordinate Chart)

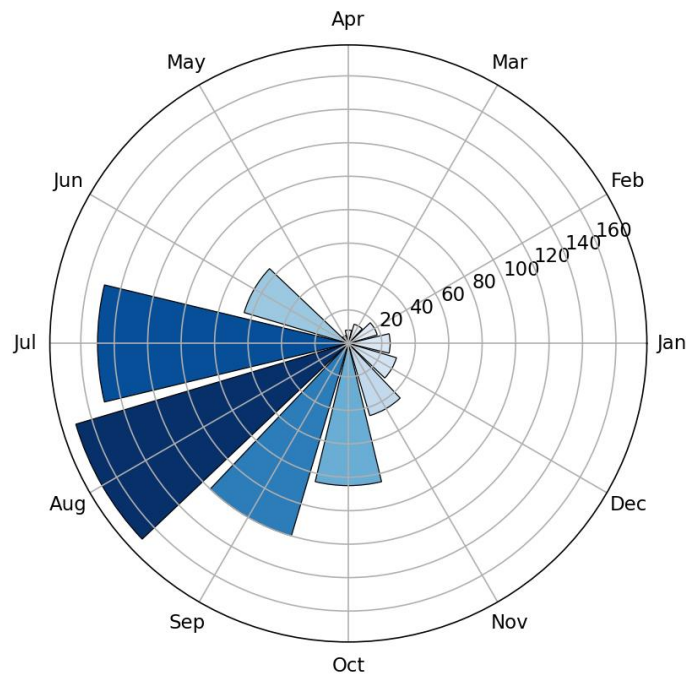


Figure 2b: Polar coordinate chart of annual rainfall

Comparison: Cartesian vs Polar Coordinates

- **Cartesian (Temperature) — Advantages:** Precise value comparison is easy since equal vertical distances always mean equal differences; trend direction (rise/fall across months) is immediately visible; most people read line charts intuitively without extra explanation.
- **Cartesian — Limitation:** A straight line does not visually convey that December loops back to January, so the cyclical nature of the seasonal calendar is not represented.
- **Polar (Rainfall) — Advantages:** Because the months are arranged in a circle, the chart naturally reflects the cyclical, repeating nature of a calendar year — useful for spotting seasonal monsoon peaks (Jul–Aug) at a glance.
- **Polar — Limitation:** Bar lengths are harder to compare precisely because they radiate outward and angular distortion grows further from the center, and unfamiliar viewers often misjudge magnitude compared with a simple bar chart.
- **Overall:** Cartesian charts suit continuous, linear trends (temperature drift), while polar charts suit cyclical/periodic phenomena (rainfall across a repeating annual calendar) — so the two were paired deliberately with the variable whose structure they represent best.

Question 3 — Business Analytics & Marketing

Interdisciplinary Integration: Marketing + Business Intelligence + Visualization

R Program (ggplot2)

```
library(ggplot2)

sales_df <- data.frame(
  Product = c("A", "B", "C", "D", "E"),
  Sales = c(210, 185, 260, 145, 120),
  Profit = c(18, 14, 25, 12, 10),
  MarketShare = c(24, 20, 28, 15, 13)
)

sales_df$Highlight <- ifelse(
  sales_df$MarketShare == max(sales_df$MarketShare),
  "Highest Market Share", "Other")

ggplot(sales_df, aes(x = Product, y = Sales, fill = Profit)) +
  geom_col(color = "black") +
  geom_col(data = subset(sales_df, Highlight == "Highest Market Share"),
    aes(x = Product, y = Sales), fill = "#2ecc71", color = "black") +
  scale_fill_gradient(low = "#ffffb2", high = "#bd0026", name = "Profit (%)") +
  labs(title = "Sales by Product",
    subtitle = "Green bar = highest Market Share",
    x = "Product", y = "Sales (₹ Lakhs)") +
  theme_minimal(base_size = 13)
```

Output

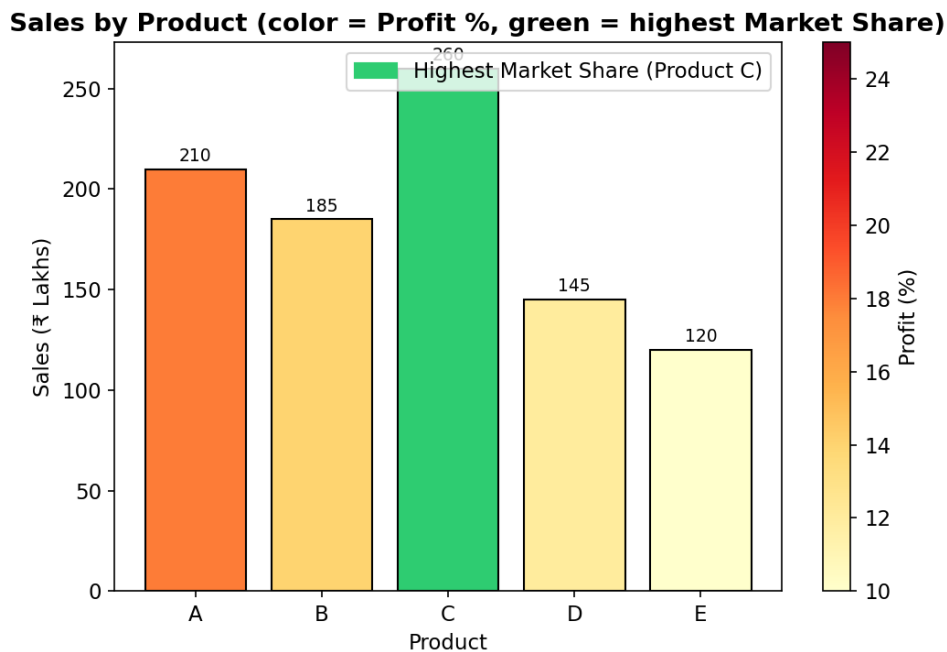


Figure 3: Sales by product; color intensity encodes Profit %, green highlights the product with the highest Market Share (Product C)

Interpretation

- **Bar height (Sales):** Bar height gives an immediate, accurately-scaled comparison of Sales across products — the most important KPI for a business review — since length is the most precisely perceived visual channel.
- **Sequential color scale (Profit %):** A sequential yellow-to-red scale is used for Profit % because profit is a continuous, ordered quantity; sequential palettes (unlike diverging or categorical ones) correctly imply 'low to high' without a false midpoint, letting a manager scan for the most profitable products at a glance.
- **Highlight color (Market Share):** Product C is picked out in a contrasting green because it breaks from the profit gradient — using a distinct hue rather than a shade of the same ramp ensures the top Market Share performer cannot be confused with a merely high-profit bar.
- **Why this improves decision-making:** This encoding lets a decision-maker answer three questions from one chart — which product sells the most, which is most profitable, and which dominates the market — supporting faster, evidence-based marketing and investment decisions.

Question 4 — Transportation Engineering

Interdisciplinary Integration: Civil Engineering + GIS + Transportation Analytics

R Program (ggplot2)

```
library(ggplot2)

traffic <- data.frame(
  Direction = factor(c("North", "North-East", "East", "South-East",
    "South", "South-West", "West", "North-West"),
    levels = c("North", "North-East", "East", "South-East",
    "South", "South-West", "West", "North-West")),
  Vehicles = c(980, 760, 1240, 890, 1420, 1130, 860, 720)
)

ggplot(traffic, aes(x = Direction, y = Vehicles, fill = Vehicles)) +
  geom_col(width = 0.9, color = "black") +
  scale_fill_viridis_c(option = "plasma", name = "Vehicles/hr") +
  coord_polar() +
  labs(title = "Traffic Flow by Direction", x = "", y = "Vehicles/hour") +
  theme_minimal(base_size = 13)
```

Output

Q4: Traffic Flow by Direction (Polar Coordinate Chart)

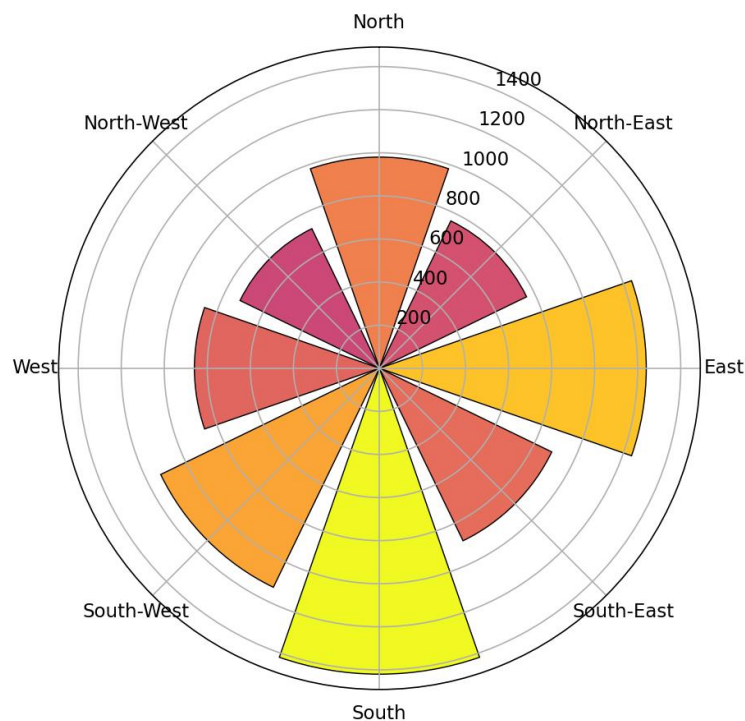


Figure 4: Polar chart of hourly traffic volume by compass direction

Interpretation

- **Why Polar suits directional data:** The eight categories (N, NE, E, SE, S, SW, W, NW) are physical compass directions radiating from a real intersection, so a polar layout preserves their true spatial/angular relationship — something a Cartesian bar chart's arbitrary left-to-right ordering cannot do.
- **Color palette for density:** A perceptually uniform sequential palette (plasma/viridis) maps low-to-high traffic volume to dark-to-bright colors, making the busiest approach (South, 1420 veh/hr) visually pop even before reading the bar length.
- **Engineering benefit:** Traffic engineers can instantly see which approaches (e.g., South and East) need extra signal time or lane capacity, mirroring how the intersection actually looks from above — supporting quicker, spatially-intuitive engineering decisions than an ordered bar chart would.

Question 5 — Environmental Sustainability Dashboard

Interdisciplinary Integration: Environmental Science + Urban Planning + Data Visualization

R Program (ggplot2 + patchwork)

```
library(ggplot2)
library(patchwork)

cities <- data.frame(
  City = c("A", "B", "C", "D", "E", "F"),
  AQI = c(48, 82, 135, 165, 72, 95),
  GreenCover = c(45, 32, 21, 18, 38, 30),
  CO2 = c(1.8, 3.5, 5.6, 7.4, 2.9, 4.2),
  Population = c(9, 15, 22, 28, 12, 18)
)

cities$AQI_Category <- cut(cities$AQI,
  breaks = c(-Inf, 50, 100, 150, Inf),
  labels = c("Good", "Moderate", "Poor", "Severe"))

cat_colors <- c(Good="#2ecc71", Moderate="#f1c40f",
  Poor="#e67e22", Severe="#e74c3c")

p1 <- ggplot(cities, aes(x = GreenCover, y = CO2,
  size = Population, color = AQI_Category)) +
  geom_point(alpha = 0.85, stroke = 0.6) +
  scale_color_manual(values = cat_colors) +
  scale_size(range = c(4, 12)) +
  labs(title = "Green Cover vs CO2 Emission",
    x = "Green Cover (%)", y = "CO2 Emission") +
  theme_minimal(base_size = 12)

p2 <- ggplot(cities, aes(x = City, y = AQI, fill = AQI_Category)) +
  geom_col(color = "black") +
  scale_fill_manual(values = cat_colors) +
  geom_hline(yintercept = c(50, 100, 150), linetype = "dashed") +
  labs(title = "AQI by City", x = "City", y = "AQI") +
  theme_minimal(base_size = 12)

p1 + p2 + plot_annotation(
  title = "Environmental Sustainability Dashboard")
```

Output

Q5: Environmental Sustainability Dashboard

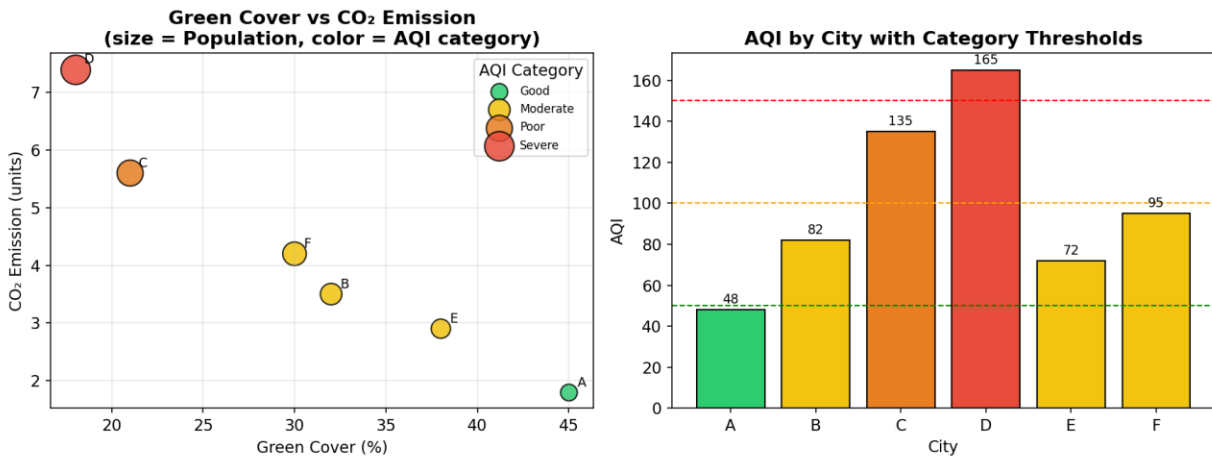


Figure 5: Dashboard — (left) Green Cover vs CO₂ Emission sized by Population, colored by AQI category; (right) AQI by city with category threshold lines

Interpretation & Design Justification

- **Aesthetic mapping:** Green Cover and CO₂ Emission are continuous variables best compared on Cartesian x/y axes to reveal the expected inverse relationship — cities with less green cover tend to emit more CO₂.
- **Scales for continuous variables:** Population is mapped to point size and AQI category to color, so a single scatter panel encodes four variables (Green Cover, CO₂, Population, AQI) at once without clutter.
- **Color palette for AQI categories:** AQI is bucketed into Good/Moderate/Poor/Severe using the standard color convention (green→yellow→orange→red), matching globally recognized air-quality index color codes so viewers interpret it instantly without a lookup table.
- **Justification for design choices:** Every choice ties directly to the decision a city-planner must make: axes show the cause-effect pair (green cover, CO₂), size shows population exposure (who is affected), color shows urgency (AQI severity), and the companion bar chart with threshold lines pinpoints exactly which cities cross into 'Poor' or 'Severe' territory.